

A Comparison of Phase Recovery Algorithms for Energy Efficient Coherent Receivers

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1 Viterbi & Viterbi

Figure 1 shows the operation of the Viterbi-Viterbi algorithm. A block of symbols $[z[k - N], z[k - N + 1], \dots, z[k], \dots, z[k + N]]$ is raised to the 4th power to remove data dependency and convolved with a maximum-likelihood (ML) filter of length $L = 2N + 1$ to compute $\hat{\theta}_{\text{ML}}[k]$. A phase unwrapper is applied to reflect the unbounded Wiener phase noise, then the unwrapped phase correction is applied to $z[k]$. Here $M = 4$ for QAM.

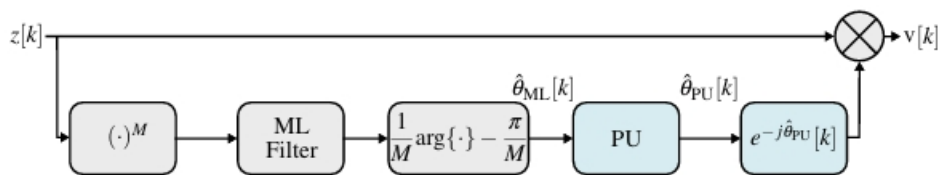


Figure 1: Viterbi & Viterbi algorithm block diagram

An energy efficient hardware implementation of this algorithm avoids multipliers where possible. The ML filter is known beforehand so the multiplications can be replaced with fixed additions and shifts. Computing the argument of the filter output and the phase correction can be implemented with CORDIC. The unwrapped phase is computed as

$$\hat{\theta}_{\text{PU}}[k] = \hat{\theta}_{\text{ML}}[k] + n[k] \frac{\pi}{2},$$

where $n[k] \in \{-1, 0, 1\}$ is calculated as

$$n[k] = \left\lfloor \frac{1}{2} + \frac{\hat{\theta}_{\text{PU}}[k - 1] - \hat{\theta}_{\text{ML}}[k]}{\pi/2} \right\rfloor.$$

The floor operation can be implemented efficiently with comparators (negligible cost) and division by $\pi/2$ can be implemented as multiplication by $2/\pi$ with fixed shifts and additions as before. However the 4th power operation requires complex multiplications which increase the energy cost. Assuming Gauss's algorithm for complex multiplication, the total cost of this algorithm per symbol in terms of real multiplications (RM), real additions (RA), and bit-shifts (BS) for n -bit fixed point precision with N_C CORDIC iterations is shown in Table 1. Note that the number of shifts and adds for multiplication

by a fixed value would be h and $h - 1$, where h is the Hamming weight, but here it is approximated as n for simplicity.

Operation	RM	RA	BS
4th Power	$9L$	$15L$	0
ML Filter	0	$2(n - 1)L$	nL
ML Phase	0	$3N_C$	$2N_C$
Phase Unwrapper	0	$3 + 2(n - 1)$	$2n$
Phase Correction	0	$3N_C$	$2N_C$
Total	$9L$	$2nL + 13L + 6N_C + 2n + 1$	$nL + 4N_C + 2n$

Table 1: Energy cost of Viterbi-Viterbi algorithm per symbol

2 Decision-Directed

The Decision-Directed algorithm avoids the 4th power operation by using a decision circuit to estimate transmitted symbols $\hat{s}[k]$. Previous estimates are then used to compute the phase correction as shown in Figure 2. Apart from the decision circuit, it is clear that any complex multiplications can be avoided by using CORDIC. The ML phase is computed as

$$\hat{\theta}_{ML}[k] = \arg \left(\sum_{i=1}^L w_{ML}[i] z[k-i] \hat{s}^*[k-i] \right) = \sum_{i=1}^L w_{ML}[i] \{ \arg(z[k-i]) - \arg(\hat{s}[k-i]) \}$$

which requires $2L$ CORDIC operations as well as shifts and additions for the ML filter.

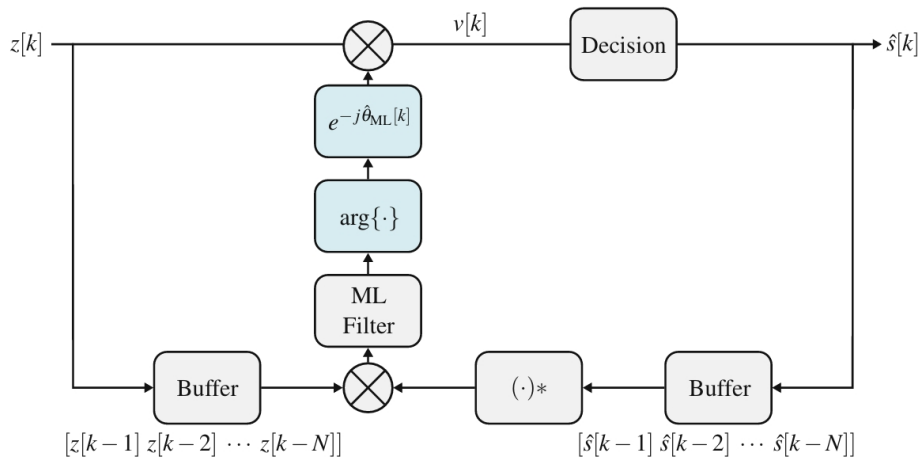


Figure 2: Decision-Directed algorithm block diagram

The decision circuit must find the constellation symbol with minimum squared distance to the corrected symbol $v[k]$. This is calculated as

$$d_m = \min_{s \in \mathcal{S}} |v[k] - \hat{s}[k]|^2$$

where $\mathcal{S}$ is the set of constellation symbols. This appears to require complex multiplications. However, in reality these distances are small and do not require high precision to

decide correctly so a look-up table (LUT) can be used. The total cost of this algorithm is shown in Table 2. Here LUT accesses are approximated to cost the same as a real addition for simplicity.

Operation	RM	RA	BS
ML Filter + Phase	0	$6LN_C + (n - 1)L + 2L - 1$	$nL + 4LN_C$
Phase Correction	0	$3N_C$	$2N_C$
Decision Circuit	0	$ \mathcal{S} $	0
Total	0	$nL + 6LN_C + L + 3N_C + \mathcal{S} - 1$	$nL + 4LN_C + 2N_C$

Table 2: Energy cost of Decision-Directed algorithm per symbol

3 Computational Cost

As each stage of CORDIC applies a left bit-shift, the accuracy achieved by N_C iterations is approximately N_C bits. Thus setting sensible values of $N_C = n = 8$, $L = 5$, and $|\mathcal{S}| = 16$ for 16-QAM, the total cost of each algorithm is shown in Table 3.

Algorithm	RM	RA	BS
Viterbi-Viterbi	45	210	88
Decision-Directed	0	324	216

Table 3: Energy cost comparison of Viterbi & Viterbi and Decision-Directed algorithms per symbol

The exact energy cost of each operation depends on how adders and multipliers are implemented, but the Decision-Directed algorithm is likely to incur a lower energy cost.

4 Block-Based Estimation

Both the Viterbi & Viterbi and Decision-Directed algorithms can be implemented in a block-based manner to reduce the energy cost. Instead of computing a phase estimate and correcting for every symbol, the estimate for one symbol can be applied to a block of B symbols. This reduces the cost of the ML filter and phase correction by a factor of B at the cost of increased phase noise variance.